Oh My Myeloma: A Rare Cause of Body Aches In A Young Adult

Lakeland Regional Health
GRADUATE MEDICAL EDUCATION

Jason Baluja, DO, Vasu Malhotra, DO, Idys Solis, MD, Graciela Luna, MD, Alaa Mohamed, MD

Dr. Kiran C. Patel College of Osteopathic Medicine
NOVA SOUTHEASTERN UNIVERSITY
NSU Florida

Lakeland Regional Health, Lakeland, Florida – ACGME program affiliated with NOVA Southeastern KCP Osteopathic Medicine

INTRODUCTION

Multiple myeloma (MM) is a malignant plasma cell disorder characterized by clonal proliferation of plasma cells in the bone marrow, leading to overproduction of monoclonal immunoglobulins and subsequent end-organ damage.

- MM accounts for 1-2% of all cancer diagnosis in the United States
- Nearly 200,00 people living with multiple myeloma in 2022.
- MM predominantly affects older populations, with a median age at onset is 69 years.
- Less than 2% of cases are diagnosed under the age of 40.

Patient Introduction

The patient is a 36-year-old male with retrolisthesis of L5 on S1 and obesity class I who presented to the Emergency Department (ED) for 8 months of progressive fatigue and diffuse body aches.

- Patient reported associated night sweats, decreased appetite, and unintentional weight loss over several months.
- Outpatient neurosurgery evaluation for the back pain, recommended conservative management.
- Presented to another ED for similar symptoms approximately one week prior and was recommended to follow up with his primary care physician. A CT scan of his pelvis done at that time for his reported left leg pain was read as unremarkable.

Work-Up And Treatment

Initial laboratory analysis was concerning for hematological malignancy as seen in Table 1, prompting further work-up for MM in the setting of the patient's symptoms and immunofixation results.

- CT imaging (Figures 1 and 2) without contrast revealed numerous radiolucent bone lesions throughout multiple axial regions including a 2 cm lesion on the left femoral neck
- Bone marrow aspiration biopsy demonstrated 70-80% of the cellularity as plasma cells
- Plasma Cell Myeloma FISH analysis revealed monosomy 13 and hyperdiploidy gains of chromosomes 5, 9, 15.
- After discharge, patient was initiated on chemotherapy with Daratumumab, Lenalidomide, Bortezomib, and Dexamethasone to which he responded well.

Laboratory Test	Result	Reference Range
Hemoglobin	9.5 g/dL	13.5–17.5 g/dL
White Blood Cell Count	1.92 × 10^3/µL	4.0–11.0 × 10^3/µL
Total Serum Protein	10 g/dL	6.0–8.3 g/dL
Albumin	2.82 g/dL	3.5–5.0 g/dL
Lactate Dehydrogenase (LDH)	320 U/L	140–280 U/L
Uric Acid	10.2 mg/dL	3.5–7.2 mg/dL
Serum Calcium	9.3 mg/dL	8.5–10.5 mg/dL
Free Kappa Light Chain	1657 mg/L	3.3–19.4 mg/L
Free Lambda Light Chain	10.9 mg/L	5.7–26.3 mg/L
Kappa/Lambda Ratio	152	0.26–1.65
Serum IgG	6098 mg/dL	700–1600 mg/dL

Table 1. Initial laboratory values

Imaging

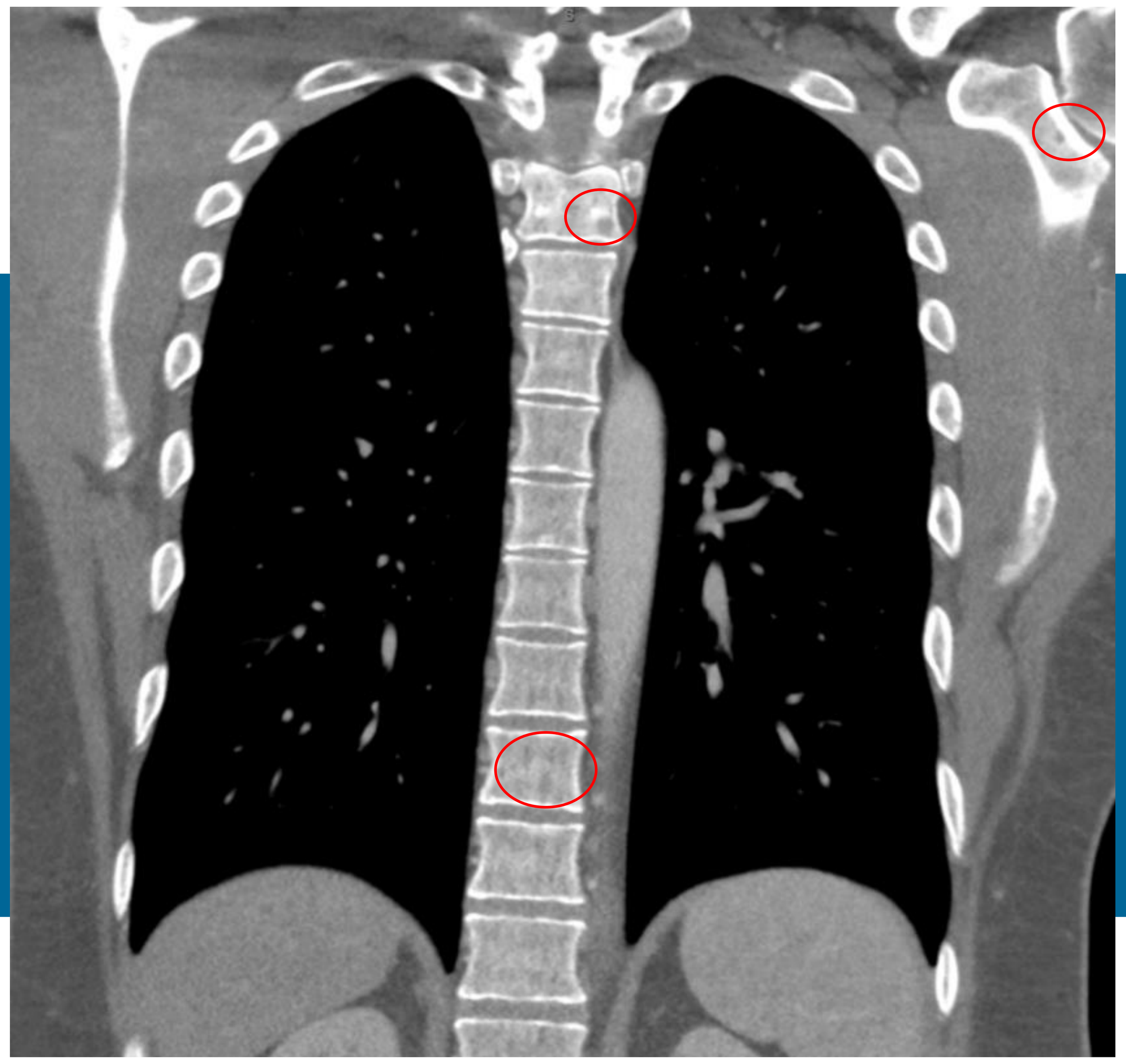

Figure 1: CT of the chest without contrast

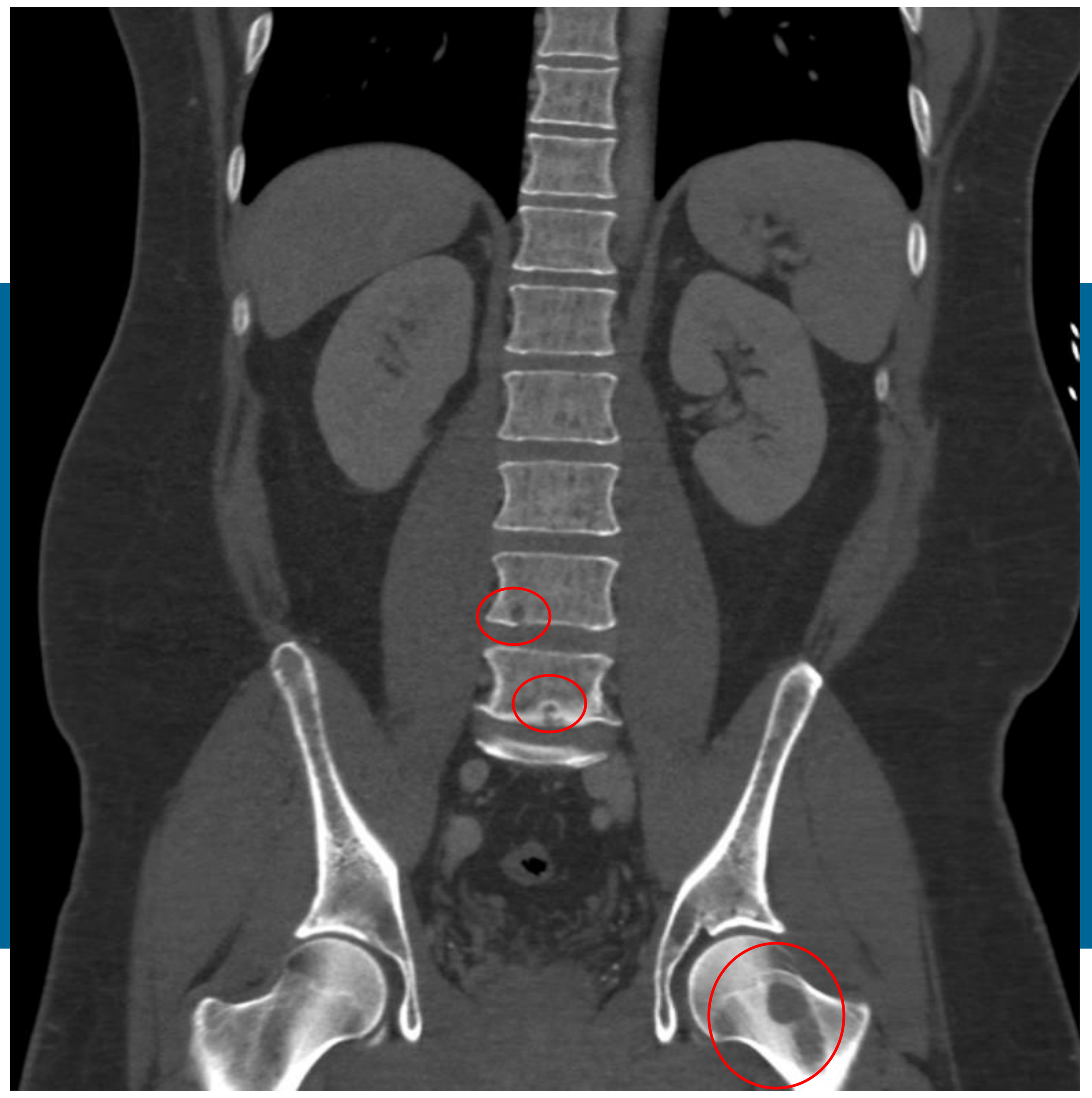

Figure 2: CT of the abdomen and pelvis without contrast

DISCUSSION

Here we presented a case of a 36-year-old male who was diagnosed with multiple myeloma. The rarity of MM in this age population allowed for several key learning opportunities.

- Bone lesions not spotted on previous imaging likely due to their subtlety and low clinical suspicion
- Young adults may have higher incidence of bone lesions and high-risk cytogenic abnormalities
- Though a classic finding, hypercalcemia is present in only 13-20% of patients at the time of diagnosis
- MM is not often a differential diagnosis that is considered in the younger population which may result in a delay in diagnosis and therefore treatment of this aggressive condition

Goals of Report

This report aims to underscore the importance of maintaining a broad differential diagnosis when evaluating unexplained chronic body aches and cytopenias in younger adults, despite the statistically low likelihood of MM in this demographic and lack of hypercalcemia.

Through this case, we contribute to the limited body of literature on early-onset multiple myeloma and emphasize the need for heightened clinical suspicion in atypical presentations so as to not delay initiation of medical care.

REFERENCES

1.
2.
3.